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Surprising sounds influence risky decision making

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Adaptive behavior depends on appropriate responses to environmental uncertainty. Incidental sensory events might simply be distracting and increase errors, but alternatively can lead to stereotyped responses despite their irrelevance. To evaluate these possibilities, we test whether task-irrelevant sensory prediction errors influence risky decision making in humans across seven experiments (total $n = 1600$). Rare auditory sequences preceding option presentation systematically increase risk taking and decrease choice perseveration (i.e., increased tendency to switch away from previously chosen options). The risk-taking and perseveration effects are dissociable by manipulating auditory statistics: when rare sequences end on standard tones, including when rare sequences consist only of standard tones, participants are less likely to perseverate after rare sequences but not more likely to take risks. Computational modeling reveals that these effects cannot be explained by increased decision noise but can be explained by value-independent risky bias and perseveration parameters, decision biases previously linked to dopamine. Control experiments demonstrate that both surprise effects can be eliminated when tone sequences are presented in a balanced or fully predictable manner, and that surprise effects cannot be explained by erroneous beliefs. These findings suggest that incidental sounds may influence many of the decisions we make in daily life.

The ability to respond appropriately to unexpected events is crucial for survival because such events can signal potential rewards or threats. Unexpected events have been theorized to motivate exploratory behaviors, allowing humans to learn about unfamiliar stimuli[1] and detect environmental changes[2,3]. Appropriate responses to environmental uncertainty are a hallmark of adaptive behavior[4].

A common source of environmental uncertainty is sensory events. Unpredictable sensory events, or sensory prediction errors, are salient and can attract attention[5]. In noisy urban environments, these sources of environmental uncertainty are often irrelevant for one's goals. Unpredictable irrelevant sensory information can lead to impaired performance in decision-making tasks such as with incidental visual distractors[6], changes in visual salience[7], acoustic frequency[8], and stimulus loudness[9]. While these effects could be explained by attentional lapses or increased noise in the decision process, sensory prediction errors could alternatively elicit systematic stereotyped responses, fixed or repetitive behaviors without obvious adaptive function[10,11]. In particular, stereotyped responses might include uncertainty seeking and option switching, consistent with the uncertainty reduction responses to task-relevant uncertainty observed in learning paradigms[12–14]. However, such responses are often adaptive in learning paradigms. It is unknown whether stereotyped responses to environmental uncertainty are present even when any such responses are not adaptive.

To test whether sensory prediction errors systematically influence risky decision making, we interleave an established risky decision-making paradigm[15–18] with auditory stimuli that previous studies[19,20] have shown elicit multiple forms of surprise. Decision making under risk is well studied with established theoretical models able to account

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for choices between safe and risky options[21–27]. Importantly, the value of potential options in risky decision-making tasks can be parametrically varied in a manner that is independent and unpredictable from trial to trial. This permits any influence of incidental sensory prediction errors on choice to be separated from the effect of changing option values, avoiding confounds that arise in tasks that require learning. Auditory stimuli are easily manipulated, statistically well defined, and amenable to eliciting multiple forms of surprise with dissociable neurophysiological correlates[28–30]. Consistent with its use in this literature, we use the term "surprise" to refer to sensory prediction errors, which entails expectation violations according to environmental statistics[31,32]. In this literature, use of this term does not imply that individuals find stimuli subjectively surprising, only that stimuli are unpredictable.

By design, the auditory stimuli in our experiments do not carry information about the probability of risky options leading to wins or losses, so making decisions based on these stimuli cannot enhance future decision making. One possibility is that incidental sounds do not have a systematic influence on behavior. Alternatively, incidental sounds could increase attentional lapses or increase decision noise, potentially consistent with previous studies suggesting performance deficits associated with irrelevant auditory noise[6–9]. A third possibility is that sensory prediction errors will introduce systematic stereotyped responses, such as influences on risk taking and perseveration. Several studies have reported increased exploration in response to task-relevant uncertainty in human learning tasks[12,13]. Systematic stereotyped responses would align with converging evidence from human and animal models showing that dopamine and norepinephrine might relate to the link between surprise and adaptive behavior. Phasic changes in arousal following administration of a noradrenergic drug increased risk taking in rodents when salient audiovisual cues were paired with outcomes[33]. A growing body of evidence has also shown that dopamine neurons respond not only to rewards but also to value-neutral sensory information like unexpected changes in stimulus identity[34–37], highlighting a link between sensory surprise and dopamine[38,39]. Endogenous fluctuations of dopaminergic midbrain activity preceding stimulus onset[40] and direct D_1 receptor stimulation[41] predicted risk taking. Notably, pharmacologically boosting dopamine increased value-independent risk taking[42,43], an effect that can be accounted for with normative basal ganglia models[44]. Pharmacologically boosting dopamine also decreases choice perseveration in learning paradigms in a manner that does not depend on rewards[45,46], which can be captured in policy compression accounts of perseveration[47]. Informed by this neurobiological and computational modeling literature, we predicted that incidental sounds would increase risk taking and decrease choice perseveration.

In each trial of our paradigm, participants chose between a risky and a safe option, with risky options having equal probabilities of winning or losing (Fig. 1; see "Methods"). Risky outcomes, if chosen, were revealed after a 1.3 s delay. Immediately before option onset for every choice trial, participants listened to a six-tone auditory sequence in which 75% of options were presented after one common sequence ("Common trial") and 25% of options were presented after a rare sequence with a different ending ("Rare trial"). Consistent with our predictions, we found in two experiments (total $n = 400$) that rare auditory sequences increased risk taking and decreased choice perseveration relative to common sequences. When tone sequences were equally balanced in two experiments (total $n = 400$), consequent differences in risk-taking and perseveration were eliminated. Using computational models, we found that these two effects do not depend on value and are dissociable. In two additional experiments (total $n = 400$), rare sequences that ended on standard tones did not increase risk taking but did decrease perseveration, consistent with perseveration being decreased by rare sequences but not rare sequence features (i.e., deviant tone transitions), which are instead associated with greater risk taking. These findings demonstrate that surprising sounds systematically influence human behavior in multiple ways.

Results

Sensory prediction errors increase risk taking

We first asked whether rare auditory sequences increased risk taking. In Experiment 1 ($n = 200$), participants chose the risky option in

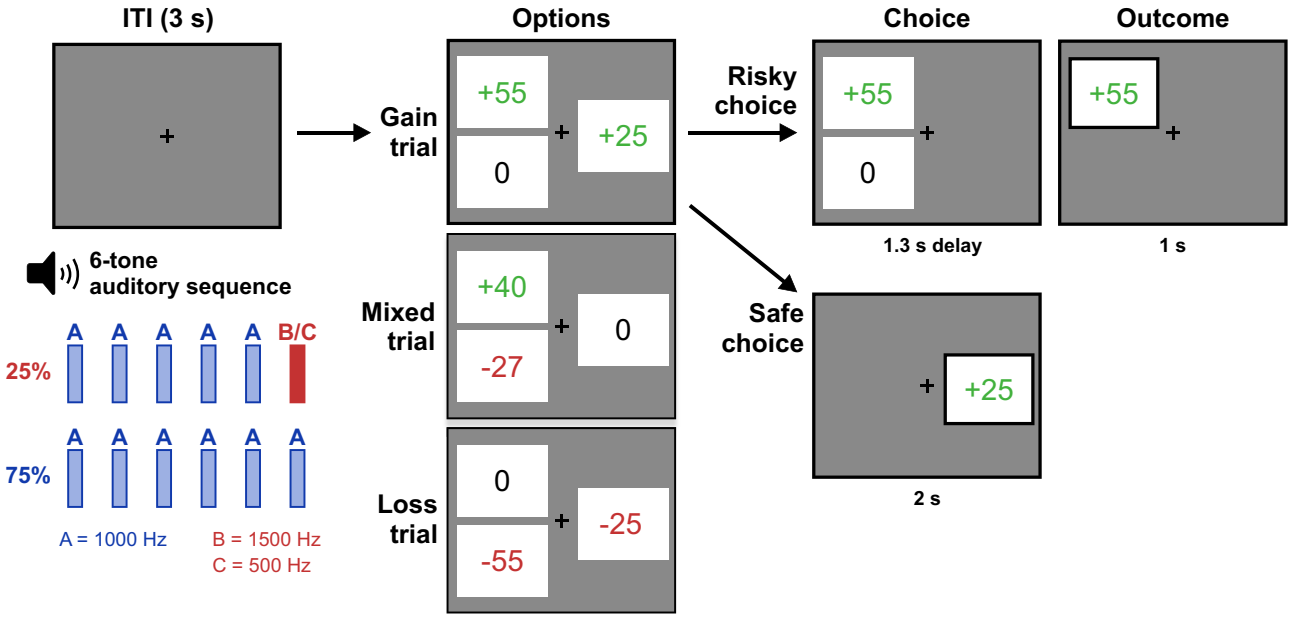

Fig. 1 | Experimental design. On every trial, participants decided whether they wanted to select a risky gamble option or a guaranteed safe option. Participants were given the goal of maximizing their cumulative point score in the task, although monetary compensation was a flat reward for completion of the experiment. Prior to option onset in every trial, six auditory tones of 100 millisecond duration were presented with 400 milliseconds between sounds. 75% of trials featured "common" sequences (AAAAAA in Experiments 1 and 2) and 25% of trials featured "rare" sequences that ended with a surprising deviant tone (AAAAAB or AAAAAC in Experiments 1 and 2). The auditory sequences were irrelevant to reward outcomes. Experiments 1 and 2 differed only in the spatial arrangement of options: in Experiment 1, the risky option was always on the left side; in Experiment 2, the spatial location of the risky option alternated every 10 trials.

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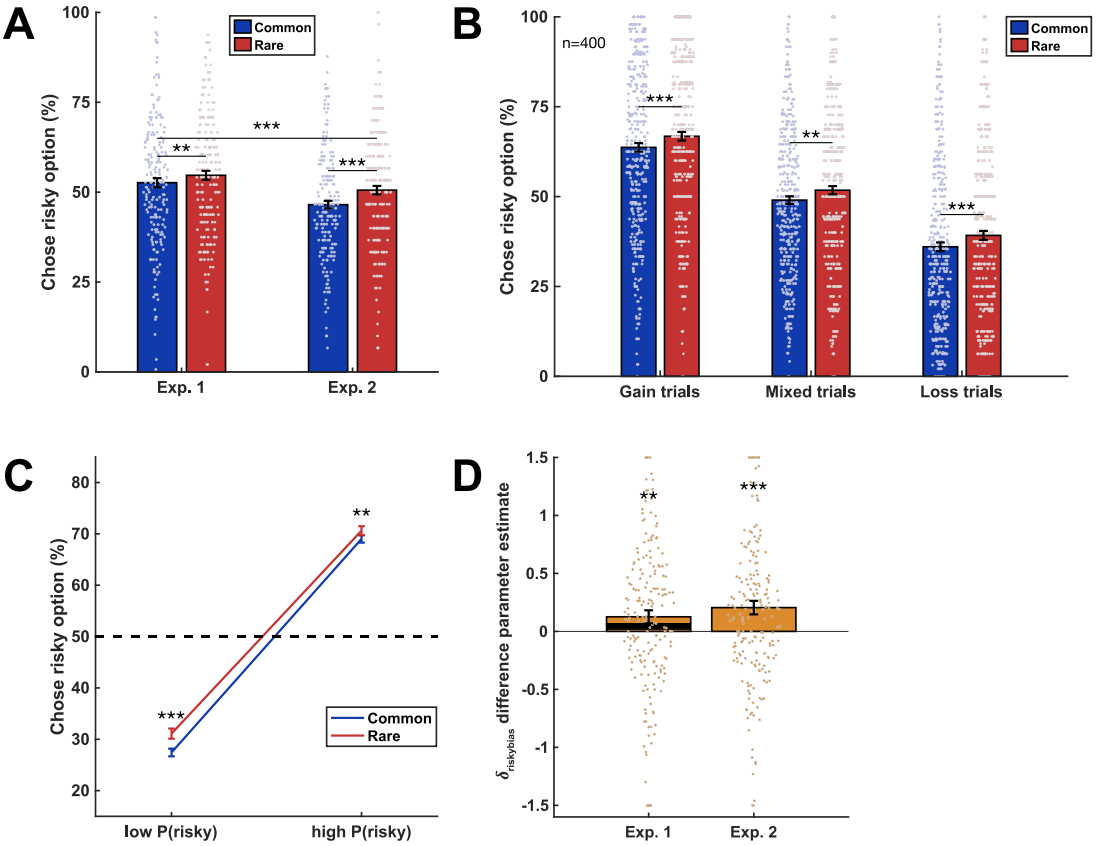

Fig. 2 | Sensory prediction errors increase risky decision making. A Average rates of risk taking for Common and Rare trials were plotted for Experiment 1 ($n = 200$ participants) and Experiment 2 ($n = 200$ participants). Participants made more risky choices on average in Rare compared to Common trials in both Experiment 1 (2.06 ± 1.09%, mean ± SEM, $p = 0.005$, two-sided Wilcoxon signed rank test) and Experiment 2 (4.03 ± 1.06%, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test). **B** For all participants in Experiments 1 and 2 combined ($n = 400$ participants), the average rates of risk taking for Common and Rare trials were compared within Gain, Mixed, and Loss trial types. Surprising sequences increase risk taking, despite variation in overall risk taking across trial types (Gain trials: $p < 0.001$, two-sided Wilcoxon signed rank test, 3.09 ± 1.01%, mean ± SEM; Mixed trials: $p = 0.010$, 2.75 ± 0.97%; Loss trials: $p < 0.001$, 3.17 ± 0.88%. **C** Trials from every participant in Experiments 1 and 2 combined ($n = 400$ participants) were split by whether the probability of risky choice predicted by the Prospect Theory model, fit individually for each participant, was below or above 50% (low *P*(risky) and high *P*(risky), respectively). Surprising sounds were associated with increased risk taking both for trials where participants tended to take risks ($p = 0.005$, two-sided Wilcoxon signed rank test, 1.61 ± 0.83% mean ± SEM) in which model-derived risk choice probability was at least 50%, and for trials where participants tended not to take risks ($p < 0.001$, two-sided Wilcoxon signed rank test, 3.67 ± 0.88%, mean ± SEM) in which the model-derived risk choice probability was less than 50%. **D** For each participant in Experiments 1 and 2 (each $n = 200$ participants), choices were fit using a parametric decision model based on Prospect Theory with a Risky Bias parameter and a $\delta_{riskybias}$ difference parameter in Rare trials to capture any change in the Risky Bias parameter in Rare relative to Common trials. The $\delta_{riskybias}$ difference parameter estimated for each participant was positive in both experiments on average, capturing increased risk taking on Rare relative to Common trials (Experiment 1: 0.125 ± 0.052, mean ± SEM, $p = 0.004$, two-sided Wilcoxon signed rank test; Experiment 2: 0.195 ± 0.056, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test). For visualization purposes, all points beyond the (−1.5,1.5) *y*-axis limits were plotted at the upper and lower boundaries. Error bars indicate SEM, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

53.16 ± 1.17% (mean ± SEM) of trials. On average, participants chose the risky option 2.06 ± 1.09% (mean ± SEM) more on Rare trials, which ended on a deviant tone, compared to Common trials ($p = 0.005$, two-sided Wilcoxon signed rank test; Fig. 2A). In Experiment 2 ($n = 200$), when the left-right spatial arrangement of options alternated every 10 trials, participants chose the risky option in 47.54 ± 0.98% of trials, less than in Experiment 1 ($p < 0.001$, two-sided Wilcoxon rank sum test). Moreover, median reaction times in Experiment 2 (1.32 ± 0.03 s, mean ± SEM) were slower compared to Experiment 1 (1.26 ± 0.03 s, mean ± SEM; $p = 0.028$, two-sided Wilcoxon rank sum test), consistent with a modest increase in cognitive demand for a non-static spatial arrangement of options. In Experiment 2, we replicated the increased risk-taking effect, finding that participants chose the risky option 4.03 ± 1.06% (mean ± SEM) more on Rare compared to Common trials ($p < 0.001$, two-sided Wilcoxon signed rank test; Fig. 2A).

To verify that the risk-taking effect of sensory surprises in these two experiments are not simply due to risky choices on Rare trials being more frequently rewarded than on Common trials, we computed how often participants won risky options in Rare trials (51.77 ± 0.61%, mean ± SEM) compared to in Common trials (50.55 ± 0.39%, mean ± SEM). Across Experiments 1 and 2, the probabilities of winning risky option outcomes were not significantly different for Rare trials relative to Common trials (1.21 ± 0.74%, mean ± SEM, $p = 0.099$, two-sided Wilcoxon signed rank test, Bayes Factor $BF_{01} = 4.677$). Even considering the subset of participants who received fewer rewards on Rare trials than Common trials by chance ($n = 188$), these participants chose the risky option 5.05 ± 1.36% more on Rare trials compared to Common trials ($p < 0.001$, two-sided Wilcoxon signed rank test). We combined Experiment 1 and 2 datasets ($n = 400$ participants) for further analyses of subsets of trials. In this combined sample, we observed greater risk taking after rare compared to Common trials, irrespective of trial type (Gain trials: $p < 0.001$, two-sided Wilcoxon signed rank test, 3.09 ± 1.01%, mean ± SEM; Mixed trials: $p = 0.010$, 2.75 ± 0.97%; Loss trials: $p < 0.001$, 3.17 ± 0.88%; Fig. 2B). Thus, the risk-taking effect does not depend on option valence, as it is present both in Gain trials where people tend

to take risks (64.47 ± 1.12%, mean ± SEM) and in Loss trials where they tend not to (36.84 ± 1.16%, mean ± SEM).

We fitted choice behavior using a Maximum Likelihood parameter estimation procedure in individual participants with a parametric decision model based on Prospect Theory, an established model which operationalizes concepts of risk and loss aversion in economic decision making[17,21]. Using computational modeling, we could precisely quantify the influences of sensory surprise on the decisions each participant made between risky and safe options, using Prospect Theory to account for different participants having different but stable overall risk attitudes.

This 4-parameter model (SI Methods), which included parameters for loss aversion, risk aversion in gains and losses, and choice stochasticity (i.e., decision noise), fit the behavior well in both Experiment 1 (pseudo-r^2 = 0.208 ± 0.011, mean ± SEM) and Experiment 2 (pseudo-r^2 = 0.223 ± 0.012, mean ± SEM). On average, participants had loss aversion coefficients greater than 1, consistent with the presence of loss aversion in our population (Experiment 1: λ = 2.394 ± 0.136, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test; Experiment 2: λ = 1.995 ± 0.120, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test; Supplementary Table 1). To test whether the increased risk taking depended on the subjective value of risk taking which varied across trials, we binned participant trials in both experiments according to whether the model-derived probability of choosing the risky option was below or above 50% for the subset of participants (n = 325) who had at least 25% of trials in each bin. The risk-taking effect was present (Fig. 2C) both for trials where participants tended to take risks (p = 0.005, two-sided Wilcoxon signed rank test, 1.61 ± 0.83% mean ± SEM) in which model-derived risk choice probability was at least 50%, and for trials where participants tended not to take risks ($p < 0.001$, two-sided Wilcoxon signed rank test, 3.67 ± 0.88%, mean ± SEM) in which the model-derived risk choice probability was less than 50%. When we compared the risk-taking effect with behavior predicted by an optimal expected value maximizing strategy, we observed that sensory surprise resulted in participants taking more risks both in situations where increased risk taking is optimal and also when it is not optimal (Supplementary Fig. 2). These results suggest that sensory surprise induces a value-independent increase in risk taking, which is sometimes but not always maladaptive.

Several studies have highlighted the importance of quantifying influences on risk taking that do not depend on the subjective value of risky and safe options, particularly in relation to potential dopaminergic influences on behavior. These influences can be accounted for by a value-independent gambling bias parameter in the softmax choice probability function[40,48–50]. To account for any value-independent influence on risk taking that cannot be captured by risk or loss aversion, we introduced a Risky Bias parameter, *riskybias*, to the softmax rule in the Prospect Theory model (Eq. (1)):

$$P_{risky}(t) = \frac{1}{1 + \exp(-[\mu(\text{Utility}_{risky}(t) - \text{Utility}_{safe}(t)) + \text{riskybias}])} \quad (1)$$

According to this model, a positive Risky Bias parameter reflects an increased tendency to choose risky options. Conversely, a negative Risky Bias parameter reflects an increased tendency to choose the safe option. Model comparison using the Akaike Information Criterion (AIC) confirmed that this Risky Bias model explained the choice data in Experiments 1 and 2 better than the Prospect Theory model (Risky Bias model: AIC = 68277; Prospect Theory model: AIC = 70961; Supplementary Table 2). The average Risky Bias parameter was lower in Experiment 2 relative to Experiment 1 (Experiment 1: 0.109 ± 0.071, mean ± SEM; Experiment 2: −0.222 ± 0.060, mean ± SEM; $p < 0.001$, two-sided Wilcoxon rank sum test), consistent with our observation that risk taking overall was reduced in Experiment 2.

We next tested whether the risk-taking effect could be explained by a change in the Risky Bias parameter. We included an additional $\delta_{riskybias}$ difference parameter in Rare trials to capture any change in Rare compared to Common trials. For comparison, we considered four alternative models that each featured a single difference parameter for one of the other four parameters in Prospect Theory (δ_μ,δ_λ,$\delta_{\alpha_{gain}}$,$\delta_{\alpha_{loss}}$). Model comparison showed that the Risky Bias Difference model, which featured a $\delta_{riskybias}$ difference parameter, best explained the data (Supplementary Table 2). Consistent with the overall pattern of increased risk taking in Rare relative to Common trials in Experiments 1 and 2 (Fig. 2A), $\delta_{riskybias}$ difference parameters were positive on average in both experiments (Experiment 1: 0.125 ± 0.052, mean ± SEM, p = 0.004, two-sided Wilcoxon signed rank test; Experiment 2: 0.195 ± 0.056, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test; Fig. 2D). While it is possible that the effects of sensory surprise on decision making could vary with subjective value of options, as suggested by the weak asymmetry in the average Rare-Common differences in risk taking for risky and non-risky trials (1.61 ± 0.84% vs. 3.69 ± 0.90%, mean ± SEM, p = 0.192, two-sided Wilcoxon signed rank test), the value-independent parameters in our Risky Bias Difference model are sufficient to recapitulate this overall pattern (Supplementary Fig. 3). Overall, these results demonstrate that surprising sounds lead to an increase in risk taking that can be explained by a single model parameter that captures change in a value-independent influence on risk taking.

Sensory prediction errors decrease choice perseveration

Next, we investigated whether surprising sounds influence how likely participants are to repeat choices. We quantified this tendency by computing the percentage of trials in which the same choice (e.g., choosing the safe option) was made on consecutive trials. In Experiment 1, participants stayed with the previously chosen option in 59.10 ± 0.72% (mean ± SEM) of trials. On average, participants stayed with their previous choice less frequently on Rare compared to Common trials ($p < 0.001$, two-sided Wilcoxon signed rank test, −3.69 ± 0.81%, mean ± SEM; Fig. 3A). In Experiment 2, which had an alternating left-right spatial arrangement of risky and safe options, participants repeated choices 54.91 ± 0.63% (mean ± SEM) of the time, less frequently compared to Experiment 1 ($p < 0.001$, two-sided Wilcoxon rank sum test). Nevertheless, in Experiment 2, participants also stayed with their previous choice less frequently on Rare compared to Common trials ($p < 0.001$, two-sided Wilcoxon signed rank test, −3.62 ± 0.89%, mean ± SEM; Fig. 3A). Thus, the effect of surprising sounds on increasing the probability of switching options was present in experiments with different overall levels of perseveration.

Irrespective of trial type, participants chose to stay with the previous option less on Rare relative to Common trials (n = 400, Gain trials: $p < 0.001$, two-sided Wilcoxon signed rank test, −3.51 ± 0.89%, mean ± SEM; Mixed trials: $p < 0.001$, −4.63 ± 0.87%; Loss trials: p = 0.006, −2.68 ± 0.82%; Fig. 3B). Thus, the perseveration effect is present both in trial types with only potential gains and in trial types with only potential losses. Next, we investigated whether the decrease in staying with previous options after rare sequences was present both on trials for which staying was or was not predicted by the Prospect Theory model. We binned trials according to whether the choice probability of staying with the previous option predicted by Prospect Theory was above or below 50%. Among participants who had at least 25% of data in both bins (n = 385), the perseveration effect was present both for trials where participants were predicted to stay with the previous option ($p < 0.001$, two-sided Wilcoxon signed rank test; −5.14 ± 0.81%, mean ± SEM), and for trials where participants were predicted not to stay with the previous option (p = 0.019, two-sided Wilcoxon signed rank test; −1.57 ± 0.73%, mean ± SEM; Fig. 3C). Thus, both between and within trial types we see that the perseveration

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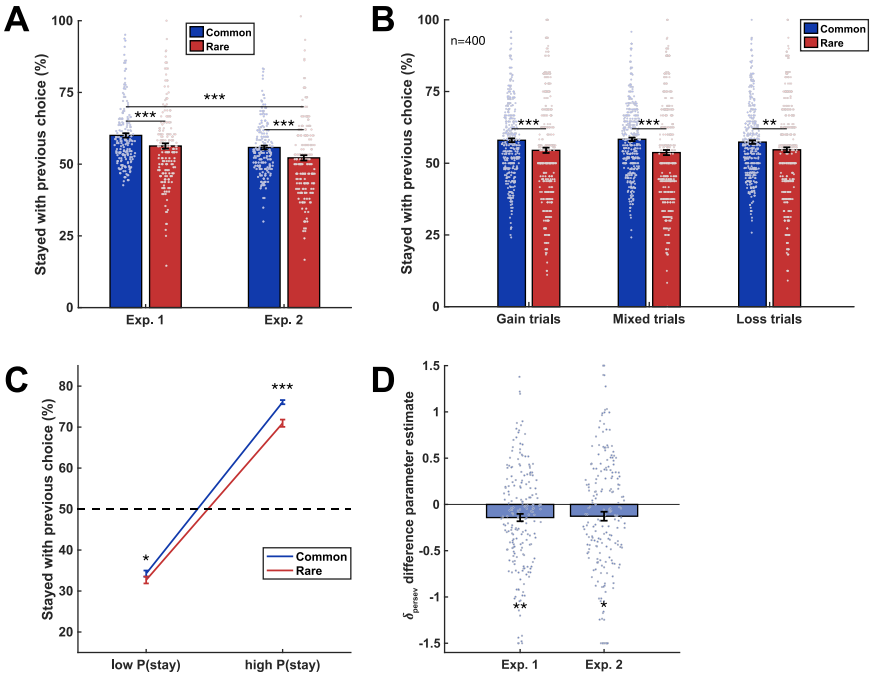

Fig. 3 | Sensory prediction errors decrease choice perseveration. A Average rates of option staying for Common and Rare trials were plotted for Experiment 1 ($n = 200$ participants) and Experiment 2 ($n = 200$ participants). Participants were less likely to repeat choices in Experiment 2 (mean = 54.91%) relative to Experiment 1 (mean = 59.10%, $p < 0.001$, two-sided Wilcoxon rank sum test). For both experiments, the probability of staying with the previous choice (i.e., the safe or risky option) was lower for Rare compared to Common trials ($p < 0.001$, two-sided Wilcoxon signed rank test). **B** For all participants in Experiments 1 and 2 combined ($n = 400$ participants), the average rates of option staying for Common and Rare trials were compared within Gain, Mixed, and Loss trial types. Option staying on Rare trials is decreased compared to Common trials irrespective of trial type (Gain trials: $p < 0.001$, two-sided Wilcoxon signed rank test, −3.51 ± 0.89%, mean ± SEM; Mixed trials: $p < 0.001$, −4.63 ± 0.87%; Loss trials: $p = 0.006$, −2.68 ± 0.82%). **C** Trials from every participant in Experiments 1 and 2 combined ($n = 400$ participants) were binned based on whether the model-predicted choice probability of staying with the previous option was above or below 50%. The perseveration effect was present both for trials where participants were predicted to stay with the previous option ($p < 0.001$, two-sided Wilcoxon signed rank test; −5.14 ± 0.81%, mean ± SEM), and for trials where participants were predicted not to stay with the previous option ($p = 0.019$, two-sided Wilcoxon signed rank test; −1.57 ± 0.73%, mean ± SEM). **D** For each participant in Experiments 1 and 2 (each $n = 200$ participants), behavior was fit with the full Risky Bias Perseveration Difference model. The average δ_{persev} difference parameter was negative in Experiment 1 (−0.149 ± 0.041, mean ± SEM, $p = 0.001$, two-sided Wilcoxon signed rank test) and Experiment 2 (−0.126 ± 0.048, mean ± SEM, $p = 0.010$, two-sided Wilcoxon signed rank test), capturing a decrease in choice perseveration on Rare trials in both experiments. For visualization purposes, all points beyond the (−1.5,1.5) y-axis limits were plotted at the upper and lower boundaries. Error bars indicate SEM, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

effect is present irrespective of whether given trials have high or low predicted option repetition probability.

We included a Perseveration parameter in the softmax equation, biasing choice according to the option selected in the previous trial (Eq. (2)). Model comparison using Akaike Information Criterion (AIC) confirmed that adding a Perseveration parameter to the Risky Bias model explained the choice data in Experiments 1 and 2 better (Risky Bias model: AIC = 68277; Risky Bias Perseveration model: AIC = 67481). The Perseveration parameter was lower in Experiment 2 relative to Experiment 1 (0.282 ± 0.031, mean ± SEM in Experiment 1; 0.129 ± 0.032, mean ± SEM, in Experiment 2; $p < 0.001$, two-sided Wilcoxon rank sum test), consistent with the observation that overall perseveration for Experiment 2 was lower than in Experiment 1:

$$P_{risky}(t) = \frac{1}{(1 + \exp(-[\mu(\text{Utility}_{risky}(t) - \text{Utility}_{safe}(t)) + p + \text{riskybias}])}$$
$$p = \text{persev} * \left(\text{Choice}_{risky}(t-1) - \text{Choice}_{safe}(t-1)\right) \quad (2)$$

We considered models with difference parameters for pairs of the six model parameters ($\delta_{\mu}, \delta_{\lambda}, \delta_{\alpha_{gain}}, \delta_{\alpha_{loss}}, \delta_{riskybias}, \delta_{persev}$). Model comparison showed that a model with $\delta_{riskybias}$ and δ_{persev} value-independent difference parameters was preferred to six models with all possible pairs of the four value-dependent difference parameters (Supplementary Table 3). In the winning Risky Bias Perseveration Difference model, $\delta_{riskybias}$ difference parameters were positive on average in both Experiment 1 (0.149 ± 0.056, mean ± SEM, $p = 0.001$, two-sided Wilcoxon signed rank test; Supplementary Table 4) and Experiment 2 (0.202 ± 0.058, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test). The average δ_{persev} difference parameter for the winning model was negative in Experiment 1 (−0.149 ± 0.041, mean ± SEM, $p = 0.001$, two-sided Wilcoxon signed rank test) and Experiment 2 (−0.126 ± 0.048, mean ± SEM, $p = 0.010$, two-sided Wilcoxon signed rank test), capturing a decrease in choice perseveration on Rare trials in both experiments. Model-derived predictions from the winning Risky Bias Perseveration Difference model successfully capture average overall risk taking and stay behavior, in addition to recapitulating the key behavioral differences of interest between Rare and Common trial conditions in the raw data (Supplementary Fig. 6). The winning model parameter recovery analysis confirmed that all model parameters including the $\delta_{riskybias}$ and δ_{persev} difference parameters were recoverable (Supplementary Fig. 1). Model recovery analyses involving data simulated by Prospect Theory, the Risky Bias Perseveration model without $\delta_{riskybias}$ and δ_{persev} difference parameters, and the full Risky Bias Perseveration Difference Model confirmed that the winning Risky Bias Perseveration Difference model was recoverable (Supplementary Fig. 4). Since the winning model features more than one difference parameter ($\delta_{riskybias}$ and δ_{persev} difference parameters), there is more than one parameter that can account for variance in behavior on Rare

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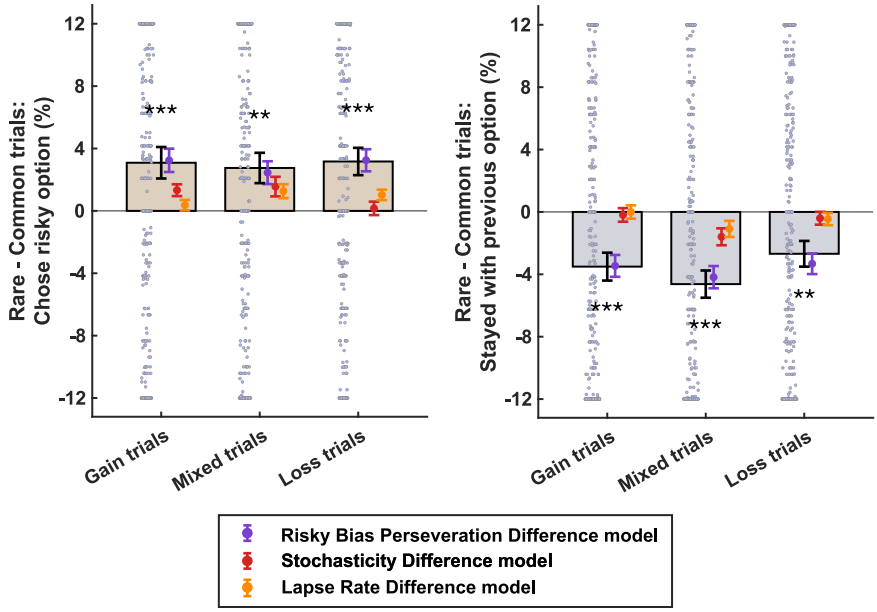

Fig. 4 | Sensory prediction error effects on risk taking and choice perseveration are not explained by a choice lapse model or a choice stochasticity model. For Experiments 1 and 2 combined ($n = 400$ participants), within-individual Rare-Common differences in the probability of risk taking (left) and staying with the previous choice (right) were computed for Gain, Mixed, and Loss trial types. Model-derived predictions are shown for the Risky Bias Perseveration Difference model, the Stochasticity Difference model, and the Lapse Rate Difference model. The Stochasticity Difference model and Lapse Rate Difference model failed to capture either risk-taking or perseveration effects. For visualization purposes, all points beyond the *y*-axis limits are plotted at the upper and lower boundaries. Error bars indicate SEM, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

trials. In addition to reporting model-agnostic raw Rare-Common differences in percentage risk taking and percentage stay rates, we also computed each of those measures after accounting for variance explained by the other effect. The model residual results were computed for all experiments and summarized in Supplementary Table 6, and all results were consistent with the full model-based analyses.

Sensory surprise effects are not explained by changes in choice stochasticity or lapse rate

We addressed several alternative explanations for the observed effects. Task-irrelevant distractor sounds have been shown to prolong reaction times and lead to lapses that reduce detection and categorization accuracy[51,52]. Thus, we considered an alternative model that used a lapse rate, which reflects a constant rate of errors independent of option values (Eq. (3)):

$$P_{\text{risky}}(t) = \frac{1-\text{lapse}}{1+\exp\left(-\left[\mu\left(\text{Utility}_{\text{risky}}(t) - \text{Utility}_{\text{safe}}(t)\right)\right]\right)} + (0.5 * \text{lapse}) \tag{3}$$

We added a δ_{lapse} difference parameter to see if the effects of sensory prediction errors observed in Experiments 1 and 2 combined could be explained by an increased lapse rate. The overall lapse rate was 0.288 ± 0.011, and the δ_{lapse} difference parameter was positive (0.050 ± 0.013, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test), consistent with sensory prediction errors increasing error rates. We also investigated whether the Lapse Rate Difference model fitted to participants in Experiments 1 and 2 could generate the same qualitative patterns in risk taking and option staying observed in the data. The Lapse Rate Difference model predictions were not consistent with the data (Fig. 4). Model comparison confirms that the full Risky Bias Perseveration Difference model is preferred to this alternative model (Supplementary Table 5).

The orienting response to novel stimuli could pull perceptual resources away from task-related processes, resulting in reduced choice consistency or sensitivity[9]. Conversely, the orienting response could increase arousal and choice consistency. Thus, we tested whether systematic influences on choice stochasticity could explain observed effects. We tested whether replacing the $\delta_{\text{riskybias}}$ and δ_{persev} difference parameters with δ_{μ} in the winning Risky Bias Perseveration Difference model could produce risk taking and option staying patterns that match the observed data. We did not observe a positive or negative effect in the δ_{μ} difference parameter (0.054 ± 0.044, mean ± SEM, $p = 0.156$, two-sided Wilcoxon signed rank test, $BF_{01} = 8.42$) that would correspond to decreased or increased choice stochasticity following sensory prediction errors. The Stochasticity Difference model simulation failed to generate the observed pattern of increased risk taking and decreased option staying across all three trial types (Fig. 4). Model comparison shows that the full Risky Bias Perseveration Difference model is preferred to this alternative Stochasticity Difference model (Supplementary Table 5). Altogether, our results show that the observed behavioral effects cannot be explained by sensory prediction errors changing choice stochasticity or lapse rate.

Perseveration and risk-taking effects are dissociable

Risky Bias and Perseveration difference parameters were only weakly correlated (Spearman $\rho = -0.119$, $p = 0.017$, $n = 400$), suggesting that they might arise from distinct processes. In Experiments 3 and 4, we asked whether the risk-taking effect might be due to hearing rare sequence features (i.e., deviant tones) and not hearing rare sequences. If deviant tones increase risk taking, rare sequences should not increase risk taking in experiments where rare sequences finish on standard rather than deviant tones (Fig. 5A).

In Experiment 3, the distribution of Rare-Common differences in risk taking was not different from zero ($-2.50 \pm 1.83\%$, mean ± SEM, $p = 0.51$, two-sided Wilcoxon signed rank test). However, rates of staying with previous options in Experiment 3 decreased ($-6.00 \pm 1.02\%$, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test), consistent with the perseveration effect observed in both Experiments 1 and 2. In the Risky Bias Perseveration Difference model, the $\delta_{\text{riskybias}}$ difference parameter was no different from zero on

average (−0.020 ± 0.091, mean ± SEM, $p = 0.913$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.40$; Fig. 5). In contrast, the δ_{persev} difference parameter was negative (−0.314 ± 0.058, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test), consistent with maintenance of the perseveration effect and selective elimination of the positive risk-taking effect for rare sequences observed in Experiments 1 and 2.

In Experiment 4, we also did not observe a positive risk-taking effect of surprise in rare trials (−3.23 ± 1.43%, mean ± SEM, $p = 0.075$, two-sided Wilcoxon signed rank test, $BF_{01} = 1.07$). Nevertheless, rare sequences decreased rates of staying with previous options in Experiment 4 (−4.27 ± 0.87%, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test). These behavioral results were mirrored in our model-based analyses: the $\delta_{riskybias}$ difference parameter was no different from zero on average (−0.067 ± 0.072, mean ± SEM, $p = 0.575$, two-sided Wilcoxon signed rank test, $BF_{01} = 11.48$; Fig. 5), whereas the δ_{persev} difference parameter was negative as predicted (−0.172 ± 0.049, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test). Altogether, $\delta_{riskybias}$ difference parameters were reduced in Experiments 1 and 2 compared to Experiments 3 and 4 ($p = 0.002$, Wilcoxon rank sum test, $n = 800$).

In Experiments 3 and 4 combined ($n = 400$), behavior for Gain, Mixed, and Loss trial types were consistent with these observations. Participants chose to stay with the previous option less on rare relative to common trials, irrespective of trial type ($n = 400$, Gain trials: $p < 0.001$, two-sided Wilcoxon signed rank test, −5.67 ± 0.95%, mean ± SEM; Mixed trials: $p < 0.001$, −5.48 ± 0.98%; Loss trials: $p = 0.006$, −4.21 ± 1.00%; Fig. 5B). However, we did not observe greater risk taking after Rare compared to Common trials in any trial type (Gain trials: −3.67 ± 1.34%, mean ± SEM, $p = 0.098$, two-sided Wilcoxon signed rank test; Mixed trials −3.47 ± 1.31%, $p = 0.010$; Loss trials: −1.42 ± 1.33%, $p = 0.49$; Fig. 5B). Interestingly, we do find some evidence for a negative effect of rare sequences on risk taking that is, in fact, predicted by our theory. However, a full test of that prediction would require a substantially larger sample given the relative frequencies of deviant tones found in rare and common sequences in our paradigm. Overall, we find that risk-taking and perseveration effects are dissociable by manipulating the statistics of auditory sequences in the task. Using model-based and model-independent analyses, we find evidence in all four experiments that rare sequences decrease perseveration, irrespective of whether they contain deviant tones.

Predictable sequences eliminate perseveration and risk-taking effects

We conducted two additional experiments, which presented each tone sequence with equal frequency (50% AAAAAB, 50% AAAAAA). In these experiments, neither sequence was labeled as Rare or Common. In Experiment 5, AAAAAB and AAAAAA tone sequences alternated deterministically across trials. In Experiment 6, AAAAAB and AAAAAA tone sequences were randomly shuffled such that no sequence was rarer than the other overall. Model fit quality was comparable to previous experiments (Supplementary Table 4). In the absence of Rare or Common trials in both experiments, we observed no perseveration effect in behavior. Experiment 5 did not show decreased rates of staying in AAAAAB trials relative to AAAAAA trials (−0.20 ± 0.54%, mean ± SEM, $p = 0.845$, two-sided Wilcoxon signed rank test, $BF_{01} = 11.85$) and neither did Experiment 6 (−0.40 ± 0.57%, mean ± SEM, $p = 0.341$, two-sided Wilcoxon signed rank test, $BF_{01} = 9.98$). Moreover, the δ_{persev} difference parameters were no different from zero in Experiment 5 (0.007 ± 0.040, mean ± SEM, $p = 0.914$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.53$; Fig. 6C) and Experiment 6 (0.007 ± 0.035, mean ± SEM, $p = 0.802$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.49$), suggesting that the effect relies on participants having a prediction about sequence identity that is violated.

We also observed no risk-taking effect in behavior. In Experiment 5, there was no difference in overall risk taking between AAAAAB trials and AAAAAA trials (−0.98 ± 0.96%, mean ± SEM, $p = 0.494$, two-sided Wilcoxon signed rank test, $BF_{01} = 7.55$) and this pattern was also observed in Experiment 6 (0.93 ± 1.12%, mean ± SEM, $p = 0.635$, two-sided Wilcoxon signed rank test, $BF_{01} = 8.98$). In Experiments 5 and 6 combined ($n = 400$ participants), behavior for Gain, Mixed, and Loss trial types were consistent with these patterns of null effects. Across all trial types, there was no difference in overall risk taking between AAAAAB trials and AAAAAA trials ($n = 400$, Gain trials: 0.02 ± 0.85%, mean ± SEM, $p = 0.554$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.81$; Mixed trials: 0.13 ± 0.96%, mean ± SEM, $p = 0.90$, $BF_{01} = 17.65$; Loss trials: 0.21 ± 0.88%, $p = 0.74$, $BF_{01} = 17.30$; Fig. 6B) and no difference in overall rates of staying (Gain trials: −0.61 ± 0.77%, mean ± SEM, $p = 0.42$, two-sided Wilcoxon signed rank test, $BF_{01} = 13.06$; Mixed trials (0.17 ± 0.73%, $p = 0.97$, $BF_{01} = 17.32$; Loss trials: −0.41 ± 0.70%, $p = 0.50$, $BF_{01} = 15.03$; Fig. 6B). In model-based analyses, we observed no risk-taking effect in the $\delta_{riskybias}$ difference parameters in Experiment 5 (−0.015 ± 0.053, mean ± SEM, $p = 0.932$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.13$) and Experiment 6 (0.029 ± 0.057, mean ± SEM, $p = 0.698$, two-sided Wilcoxon signed rank test, $BF_{01} = 15.70$; Fig. 6C). Here, B tones are less frequent than A tones, but are less "deviant" compared to Experiments 1 and 2 due to a four-fold increase in the prevalence of A to B transitions. The average $\delta_{riskybias}$ difference parameter for Experiment 5 was reduced compared to both Experiment 1 ($p = 0.010$, two-sided Wilcoxon rank sum test) and 2 ($p = 0.005$, two-sided Wilcoxon rank sum test). Similarly, Experiment 6 showed an attenuated risk-taking effect captured by the $\delta_{riskybias}$ difference parameter compared to Experiments 1 ($p = 0.020$, two-sided Wilcoxon rank sum test) and 2 ($p = 0.011$, two-sided Wilcoxon rank sum test).

Sensory surprise effects are not explained by erroneous beliefs

The auditory sequences in our task were reward-irrelevant by design, meaning that they did not contain information about the probability of a risky option leading to a win or a loss. Participants may nevertheless have formed superstitious beliefs about tone sequences or think they contain information when there is none. If participants believed that rare tones indicate that the win probability has increased, they might be more likely to take risks after rare tones. Asking participants directly about their beliefs allows testing whether sensory prediction errors systematically influence risk taking even in people who report believing that tone sequences do not carry any information relevant to the probability of reward.

We conducted a pre-registered study Experiment 7 ($n = 400$) to replicate the effects of sensory prediction errors on the probability that participants take risks or switch between options, and to ask whether any effect is related to reported beliefs. For this pre-registered experiment, we also removed information about current earnings in the task and paid participants based on their performance in an incentive-compatible manner. Consistent with our previous studies, the Risky Bias Perseveration Difference model parameters in this pre-registered study were consistent with the presence of both the risk-taking effect and perseveration effect of rare tone sequences. The $\delta_{riskybias}$ difference parameter was positive (0.109 ± 0.055, mean ± SEM, $p = 0.028$, two-sided Wilcoxon signed rank test; Fig. 7) and the δ_{persev} difference parameter was negative (−0.208 ± 0.040, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test), consistent with the results of Experiments 1 and 2.

We predicted in our pre-registration that increased risk taking would be present even in participants who explicitly report that they believe tone sequences are unrelated to their odds of wins from risky options. After the study, we asked participants: "In this experiment, you heard different sequences of tones. Do you think the sequences were related to the chances of winning the gambles?" 89 participants (22%) reported believing that tone sequences carried information

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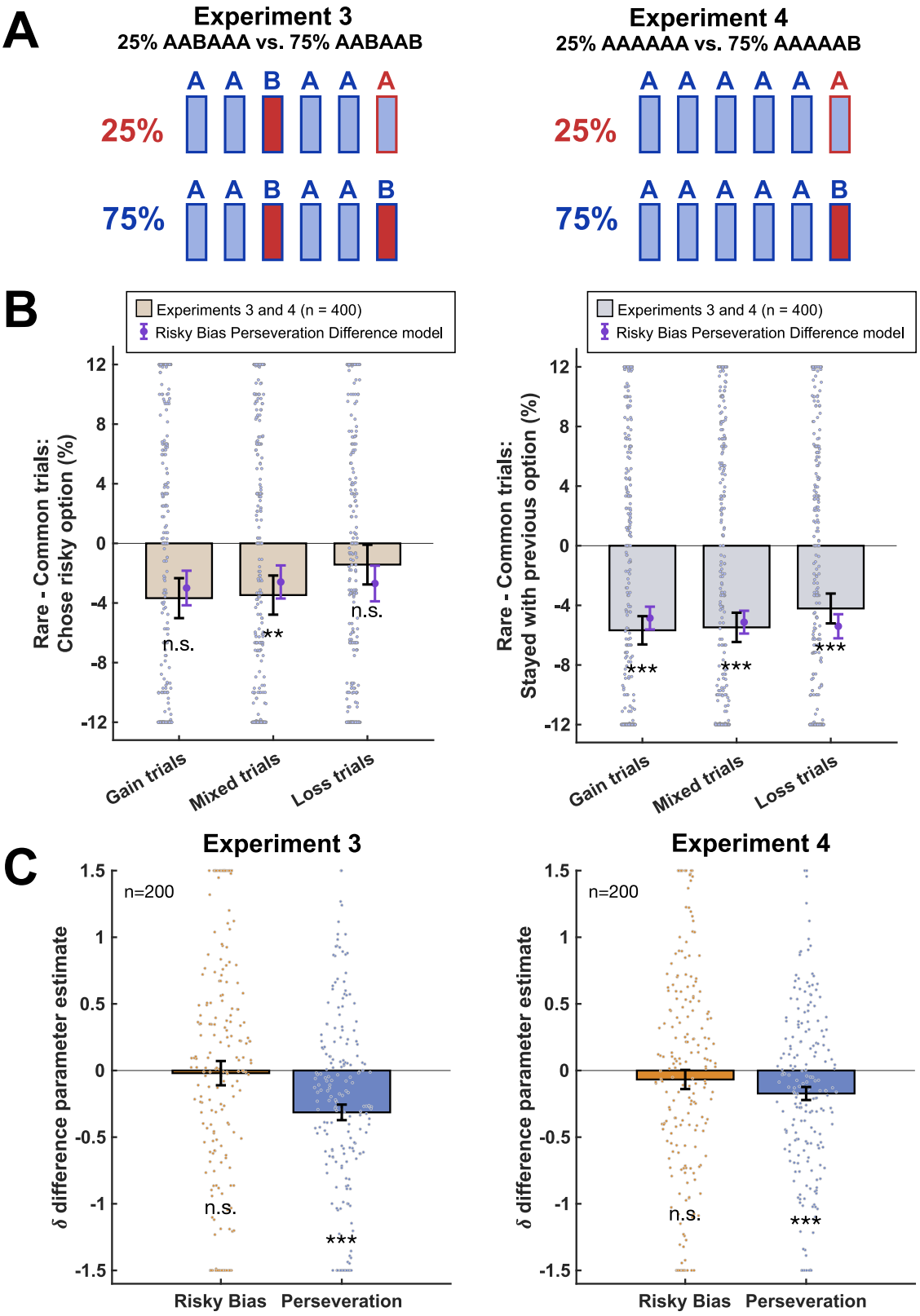

about the odds of winning. When we excluded these participants, the remaining 311 participants still showed the same risk-taking effect, with positive $\delta_{riskybias}$ difference parameters on average (0.102 ± 0.058, mean ± SEM, $p = 0.041$, two-sided Wilcoxon signed rank test; Fig. 7). In exploratory analyses, we found that the δ_{persev} difference parameters were also unchanged (-0.180 ± 0.047, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test).

Discussion

In bustling urban environments, unpredictable sensory events are common. If while walking you hear a fast-approaching car, you might quickly change directions, an appropriate response to avoiding many potential threats. It is unknown to what extent behaviors elicited by sensory surprise represent stereotyped actions, independent of goals to maximize reward and minimize threat. In seven experiments

Fig. 5 | When rare sequences end on standard tones, rare sequences decrease perseveration but do not increase risk taking. A In Experiments 3 and 4, the rare sequences ended on standard final tones (AABAAA or AAAAAA; 25% of trials), whereas the common sequences ended on deviant final tones (AABAAB or AAAAAB; 75% frequency). **B** In Experiments 3 and 4 combined ($n = 400$ participants), within-individual Rare-Common differences in the probability of risk taking (left) and staying with the previous choice (right) were computed for Gain, Mixed, and Loss trial types. Probability of staying was decreased for Rare compared to Common trials in all three trial types, consistent with Experiments 1 and 2 (Gain trials: $p < 0.001$, two-sided Wilcoxon signed rank test, −5.67 ± 0.95%, mean ± SEM; Mixed trials: $p < 0.001$, −5.48 ± 0.98%; Loss trials: $p = 0.006$, −4.21 ± 1.00%). However, unlike Experiments 1 and 2, risk taking was not increased for Rare trials in any trial type (Gain trials: −3.67 ± 1.34%, mean ± SEM, $p = 0.098$, two-sided Wilcoxon signed rank test; Mixed trials −3.47 ± 1.31%, $p = 0.010$; Loss trials: −1.42 ± 1.33%, $p = 0.49$). **C** In both Experiments 3 ($n = 200$ participants) and 4 ($n = 200$ participants), behavior for each participant was fit with the Risky Bias Perseveration Difference model, and δ_{persev} with $\delta_{riskybias}$ difference parameter estimates are displayed.δ_{persev} difference parameters were negative (Experiment 3: −0.314 ± 0.058, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test; Experiment 4: −0.172 ± 0.049, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test), consistent with decreased perseveration after Rare than Common trials as in Experiments 1 and 2. In contrast, $\delta_{riskybias}$ difference parameters did not differ significantly from zero, a selective reduction of this positive effect (Experiment 3: −0.020 ± 0.091, mean ± SEM, $p = 0.913$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.40$; Experiment 4: −0.067 ± 0.072, mean ± SEM, $p = 0.575$, two-sided Wilcoxon signed rank test, $BF_{01} = 11.48$). For visualization purposes, all points beyond the *y*-axis limits were plotted at the upper and lower boundaries. Error bars indicate SEM, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

($n = 1600$), we investigated the effects of task-irrelevant sensory prediction errors on risky decision making. In Experiments 1 and 2, we found that rare auditory sequences that ended with a deviant tone resulted in increased risk taking and decreased choice perseveration. Using a computational model based on Prospect Theory, we captured sensory prediction error effects with two value-independent parameters that modulated overall risk taking and overall choice perseveration. In Experiments 3 and 4, we found that when rare sequences ended on standard tones and common sequences ended on deviant tones, the effect of increased risk taking after rare sequences observed in Experiments 1 and 2 was selectively eliminated, while the effect of decreased perseveration after rare sequences remained intact. These results suggest that risk-taking and choice perseveration effects are unlikely to share a common mechanism. In Experiments 5 and 6, we observed that both the perseveration and the risk-taking effects were eliminated when the tone sequences were presented in a balanced or predictable manner, with both sequences presented with equal probabilities either deterministically or stochastically. Finally, in Experiment 7, we observed that our main findings replicated in participants who reported believing that tone sequences did not carry information relevant to the probabilities of winning rewards. While the phenomenology of auditory surprise itself may vary between individuals (for example, as a function of musical expertise), the observed effects on risk taking and perseveration do not depend on participants developing explicit beliefs about the behavioral relevance of deviant events. This pattern of results provides evidence against any possible mechanism that requires explicit beliefs about how surprise and behavior should be related. Overall, the consistency of both surprise effects across many different situations (e.g., with and without potential rewards, where participants tend to stay or tend to switch, where participants tend to take risks or not to take risks, and in situations where they report that surprise has no relevance to winning rewards) strongly suggests that surprise effects on behavior do not depend on the value of available options.

Surprise effects on behavior cannot be simply explained as changes in choice stochasticity. If surprising sounds make behavior less random because it enhances attention, then participants should take more risks on trials where the risky option has the higher subjective value and take fewer risks on trials where the safe option has the higher subjective value, making choices more congruent with value-based preferences. Alternatively, if surprising sounds increase choice randomness due to distraction, then the opposite effect should be observed and choices should be shifted toward indifference between risky and safe options. Contrary to both predictions, we observed that risk taking is increased on Rare trials both for trials where participants usually take risks, and for trials where participants usually do not take risks. We observed that the perseveration effect was present both for trials where participants are expected to stay with the previous option, and for trials where participants are expected not to stay with the previous option. Qualitative and quantitative model comparison shows that a Risky Bias Perseveration Difference model that captures sensory prediction error effects with only value-independent risk-taking and perseveration parameters is preferred to models that capture only enhanced attention, distraction, or choice lapses.

The pattern of results is consistent with the perseveration effect arising from recognition of a surprising target sequence. Choice perseveration was decreased after rare sequences when they were presented in 25% of trials in Experiments 1–4 and 7. The A tone was more common than the deviant B tone in all seven experiments, but the perseveration effect did not depend on the frequency of B tones: the perseveration effect was present despite wide variation in B tone frequency in Experiments 1–4 (Experiments 1, 2, and 7: 2.1%; Experiment 3: 29.2%; Experiment 4: 12.5%), and absent in control experiments with intermediate B tone frequency (Experiments 5 and 6: 8.3%). Most strikingly, the rare sequences that led to decreased perseveration in Experiment 4 contained only standard A tones and contained no deviant B tones at all.

The positive risk-taking effect cannot arise from recognition of a surprising target sequence as this effect was absent in Experiments 3 and 4, despite strong perseveration effects. This pattern of results is consistent with the risk-taking effect depending on the rarity of lower-level sequence features like deviant B tones. Another possibility is that it depends on local tone-to-tone transitions (e.g., hearing a B tone after an A tone), which have similar statistics to B tone deviance in our paradigms. In our theory, the rarer the deviant tone, the greater its effect on increasing risk taking. Experiments 3 and 4, which switched the position of the deviant B tone ending to occur on common instead of rare sequences, should flip the sign of the positive risk-taking effect. However, the deviant B tone occurred only 2.1% of the time in Experiments 1 and 2, where we observed significant risk-taking effects. Deviant B tones were much more frequent in experiments that did not show risk taking differences between Rare and Common trials (Experiment 3: 29.2%; Experiment 4: 12.5%; Experiments 5 and 6: 8.3% in the sequences with the rarer B tones). The lower overall rarity of the deviant tones in Experiments 3 and 4 means that a much larger sample size would be required to detect a negative risk-taking effect compared to the sample required to detect the reduction in the effect observed from Experiments 1 and 2 to 3 and 4.

Together, Experiments 3 and 4 highlight that risk taking and perseveration responses to sensory surprise could potentially be linked to dissociable neurobiological mechanisms that are manipulable by the auditory statistics of the environment. Sensory surprises increase phasic dopamine responses[34] and distributed prefrontal electrophysiological signals differentiate between local and global auditory surprise in humans[19]. Using dopamine antagonists, selective pharmacological inactivation of prefrontal D1 projections to the nucleus accumbens modulate reward seeking and selective inactivation of prefrontal D2 projections to the amygdala modulate behavioral

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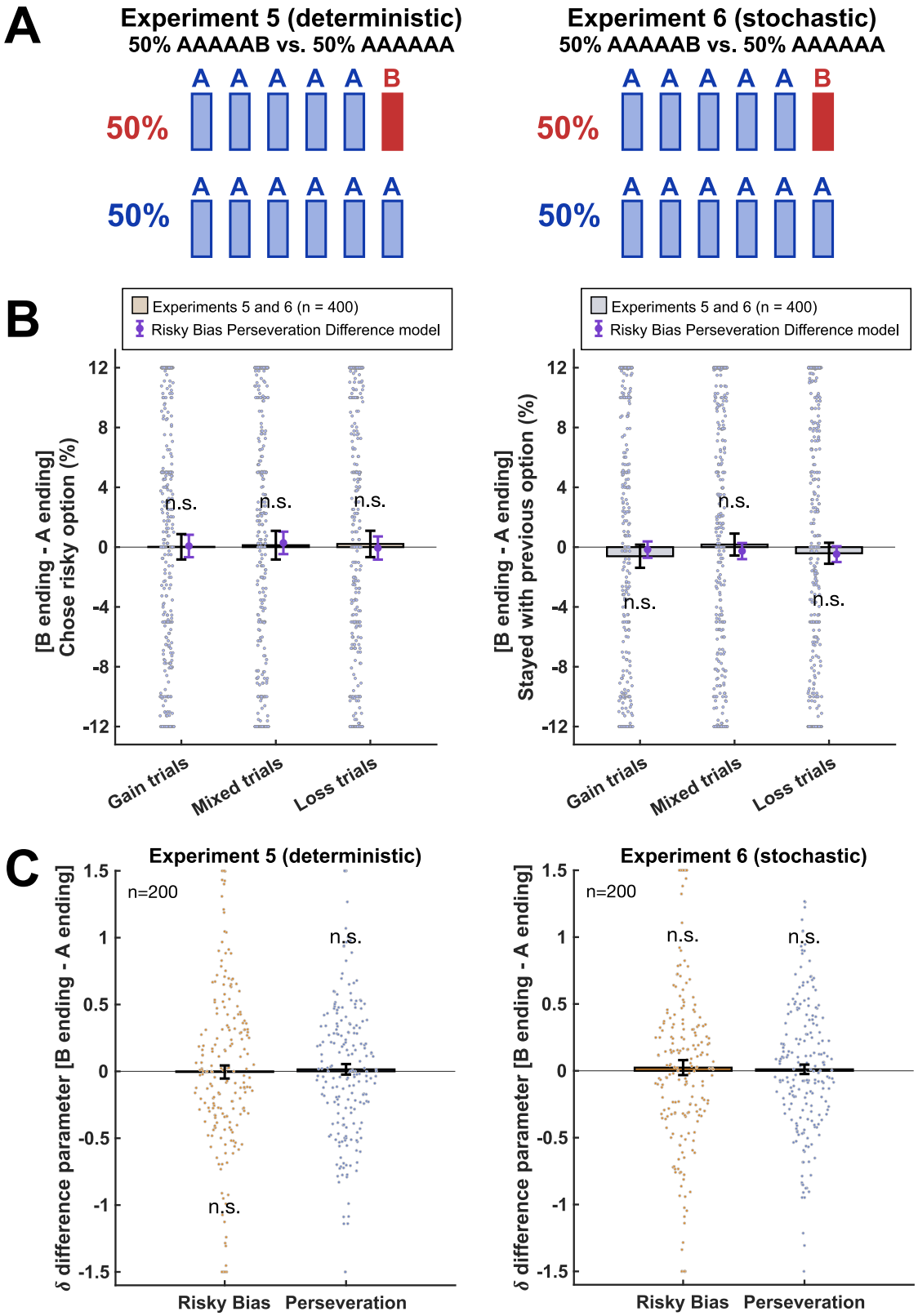

flexibility[53]. The auditory surprises delivered in our paradigm could be engaging similar underlying mechanisms and modulating the strength of dopamine-dependent prefrontal projections to the striatum and amygdala.

Pharmacological and optogenetic experiments would test our theories about the biological basis of observed local and global surprise effects on decision making. We can test this paradigm in patients with Parkinson's Disease to determine whether surprise effects are attenuated according to the extent of dopaminergic neurodegeneration and restored upon administration of levodopa. Our findings also point the way for detailed neurobiological studies in animal models to precisely test the basis of the effects we identified in humans. For example, we predict that pharmacological infusions of a dopamine receptor antagonist in rodents to disrupt D1 receptor modulation of

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Fig. 6 | Risk-taking and perseveration effects are eliminated when tone sequences are fully balanced. A In Experiments 5 and 6, both sequence types were presented with equal 50% frequency. In Experiment 5, the two sequence types alternated deterministically across trials, and in Experiment 6, the order of auditory sequences were shuffled randomly. **B** In Experiments 5 and 6 combined ($n = 400$ participants), within-individual Rare-Common differences in the probability of risk taking (left) and staying with the previous choice (right) were computed for Gain, Mixed, and Loss trial types. Risk-taking differences between AAAAAB and AAAAAA sequence types were absent, a consistent effect for all three trial types (Gain trials: 0.02 ± 0.85%, mean ± SEM, $p = 0.554$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.81$; Mixed trials: 0.13 ± 0.96%, mean ± SEM, $p = 0.90$, $BF_{01} = 17.65$; Loss trials: 0.21 ± 0.88%, $p = 0.74$, $BF_{01} = 17.30$). Staying effect differences were also absent across trial types (Gain trials: −0.61 ± 0.77%, mean ± SEM, $p = 0.42$, two-sided Wilcoxon signed rank test, $BF_{01} = 13.06$; Mixed trials (0.17 ± 0.73%, $p = 0.97$, $BF_{01} = 17.32$; Loss trials: −0.41 ± 0.70%, $p = 0.50$, $BF_{01} = 15.03$). **C** In Experiments 5 and 6 (each $n = 200$ participants), behavior was estimated with the Risky Bias Perseveration Difference model. No risk-taking effect in the $\delta_{riskybias}$ difference parameters was observed in Experiment 5 (−0.015 ± 0.053, mean ± SEM, $p = 0.932$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.13$) and Experiment 6 (0.029 ± 0.057, mean ± SEM, $p = 0.698$, two-sided Wilcoxon signed rank test, $BF_{01} = 15.70$. The δ_{persev} difference parameter estimates were also no different from zero in Experiment 5 (0.007 ± 0.040, mean ± SEM, $p = 0.914$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.53$) and Experiment 6 (0.007 ± 0.035, mean ± SEM, $p = 0.802$, two-sided Wilcoxon signed rank test, $BF_{01} = 17.49$). For visualization purposes, all points beyond the *y*-axis limits were plotted at the upper and lower boundaries. Error bars indicate SEM, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

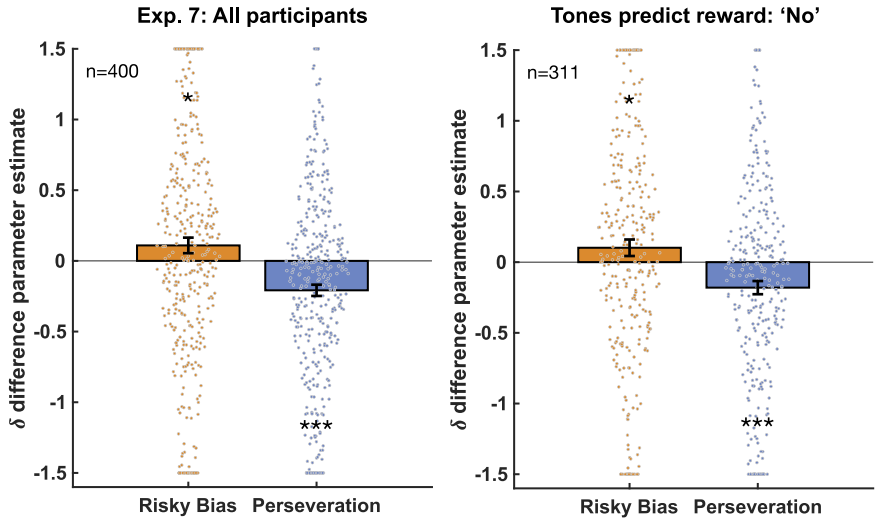

Fig. 7 | Risk-taking and perseveration effects are present in participants who report believing that tone sequences do not predict reward. In the pre-registered Experiment 7 ($n = 400$ participants), the $\delta_{riskybias}$ difference parameter was positive (0.109 ± 0.055, mean ± SEM, $p = 0.028$, two-sided Wilcoxon signed rank test) and the δ_{persev} difference parameter was negative (−0.208 ± 0.040, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test), replicating the key model-based results in Experiments 1 and 2. Even after excluding all participants (89 of 400 participants) who endorsed beliefs that tone sequences carried information relevant to the probability of reward, the remaining 311 participants still showed the same risk-taking ($\delta_{riskybias}$ parameter: 0.102 ± 0.058, mean ± SEM, $p = 0.041$, two-sided Wilcoxon signed rank test) and perseveration effects (δ_{persev} parameter: −0.180 ± 0.047, mean ± SEM, $p < 0.001$, two-sided Wilcoxon signed rank test). Error bars indicate SEM, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

prefrontal cortex neurons would result in a diminished effect of sensory surprise on risk taking. Optogenetic experiments could enable causal manipulation of sensory surprise effects at the trial level: stimulation of dopaminergic neurons in rodents during a sensory surprise would selectively enhance both effects. Silencing those neurons during a sensory surprise would eliminate them. If these predictions are satisfied, it would suggest that auditory stimuli can be used to noninvasively and selectively elicit precisely timed neuromodulator release in humans.

Physiological sensory surprise signatures have been observed in human auditory[54], visual[55], and somatosensory oddball paradigms[56,57]. In rodents, dopaminergic neuron spiking activity in VTA neurons encodes sensory prediction errors in the taste domain when equally valued juice flavor rewards are unexpectedly switched[34]. Together, these results suggest that underlying sensory prediction error mechanisms may be agnostic to sensory modality. Thus, we predict that sensory prediction error influences on behavior should also follow surprising sensory events beyond the auditory domain.

Random exploration in reinforcement-learning tasks has been explained as an adaptive response to increased overall uncertainty in the environment, computed by summing uncertainty over all options[58,59]. Exploration could also be driven by relative uncertainty between options, such that the more uncertain option is more likely to be explored under a directed exploration strategy[60]. In our experiments, increased preference for the riskier option may correspond with engagement of this directed exploration strategy and relate to the surprise associated with deviant tone transitions. In contrast, decreased perseveration may correspond with engagement of this random exploration strategy and relate to global sequence-level surprise that depends on overall environmental uncertainty. Directed and random exploration may be adaptive in response to different forms of uncertainty during reinforcement learning[12,61,62]. Multiple interacting systems are known to influence decision making[63,64] and task-relevant uncertainty or novelty may be key factors in the adaptive arbitration between decision strategies like exploration and exploitation[65–67]. In our paradigm, risky options have equal probabilities of win and loss outcomes, thus overall and relative uncertainty between safe and risky options do not change across trials. Moreover, options are randomly varied from trial to trial, unlike in many reinforcement-learning tasks. Strikingly, we find that uncertainty resulting from incidental tone sequences is sufficient to encourage decisions that could attempt to reduce uncertainty about the environment, despite these influences not being adaptive given their irrelevance. Our results suggest that task-irrelevant sensory prediction errors can be used as a tool for investigating the arbitration between decision strategies in any decision or learning paradigm.

Because surprising sensory events are everywhere in the natural world, they could have an impact on many of the decisions we make on a daily basis. This opens an avenue for field research that could investigate how the sensory surprise effects observed in this study could generalize to real-world settings. Do the sounds in a noisy cafe influence whether customers order a new item on the menu? Some real-world environments are famously noisy and full of consequential value-based decision making. Studies of professional traders suggest that the ambient noise of busy trading floors influences risk perception[68]. Our results suggest that the precise timing of surprising sounds on the trading floor influences trading decisions. Decision making in casinos would also provide a test of our theories. Our results suggest that individual decisions to hit or stand at a blackjack table would depend not only on subjective values for different combinations of cards, but also on incidental sounds common in casino environments, including for example the unpredictable sounds from nearby slot machines.

We conclude that behavioral responses to auditory surprise during decision making under uncertainty appear in at least two ways: through an increase in risk taking, and through a decrease in choice perseveration. These results also provide a demonstration for how to use noninvasive auditory stimuli to generate hypotheses about the underlying neurocognitive mechanisms of sensory prediction errors in a manner that avoids confounds related to participant choice. Moreover, our findings could offer a new way to evaluate clinical samples by examining the relationships between sensory prediction errors, behavior, and mental health. We find that surprising sounds systematically alter human behavior, identifying a commonplace yet previously unrecognized potential source of behavioral variability in everyday decision making that could have important societal consequences. The same judge, doctor, or police officer could make different decisions depending on incidental sensory events that are literally noise.

Methods

Ethical approval for this study was provided by the Yale University Institutional Review Board (protocol #2000028824).

Participants

We used the Gorilla Experiment Builder (https://gorilla.sc/) to develop our experiment. Data was collected between 24 Feb 2021 and 05 Sept 2023. Participants were recruited online through Prolific (https://www.prolific.co/). The inclusion criteria for the study were that participants spoke fluent English and were at least 18 years old. Self-reported gender but not sex information was collected for all participants. Participants provided informed consent and were paid a flat rate for study completion ($8.36/h on average). Participants were then prescreened by completing an in-browser headphone screening task[69,70]. This short task, designed to be easy for participants wearing headphones but difficult using speakers, allowed us to exclude participants not wearing headphones. This allowed us to standardize online sound presentation and mitigate concerns over sound quality. Across seven separate experiments, a total of 1600 participants passed the prescreening and finished the main task (695 female, 891 male, 14 other or preferred not to say, ages 29.0 ± 10.7 years, mean ± SD). After completing the main task, participants were administered Patient Health Questionnaire (PHQ-8) and Generalized Anxiety Disorder-7 (GAD-7) questionnaires.

Procedure

Participants were informed at the start of the experiment that they would hear common and rare tone sequences, listened to each sequence, and were told which one was common. In Experiments 1–4, participants were asked to identify the common sequence before starting the task and after completing the task. Post-task accuracy for this question was on average 95.50%. Participants were endowed with 500 points and completed a risky decision task with 192 trials (Experiment 1) or 120 trials (Experiments 2–7). On each trial, participants chose between risky and safe options worth varying amounts of points by pressing one of two computer keys. Unchosen options disappeared immediately following a choice, and chosen gambles were resolved after a 1.3-s delay. Safe and risky outcomes were revealed for 1 s. Participants were explicitly informed that all risky options had equal probability of win and loss outcomes. The experiment featured three possible types of trials: (1) Gain: a sure gain or a risky option with a larger potential gain or zero; (2) Mixed: a sure option of zero or a risky option with both a potential gain and a potential loss; and (3) Loss: a sure loss or a risky option with a larger potential loss or zero. All choice trials were self-paced, with median reaction times of 1.34 s across all seven experiments on average. In Experiments 1–6, current earnings in the task were displayed at all times. During the 3-s inter-trial interval before the onset of options for every choice trial, participants viewed a fixation cross and listened to a reward-irrelevant six-tone auditory sequence. 75% of trials featured "common" sequences, and 25% of trials featured "rare" sequences with a different ending. Each sequence was composed of six 100-millisecond pure sinusoidal tones played 400 milliseconds apart, of three possible frequencies: 1000 Hz tones (A), 1500 Hz (B), and 500 Hz (C). In Experiments 5 and 6, two sequences appeared with 50% frequency. In Experiments 1 and 3–7, the spatial position of risky and safe options was fixed. Experiment 2 differed from Experiment 1 in reversing the spatial positions of risky and safe options every 10 trials. Statistical analysis was conducted using the MATLAB statistics toolbox. Bayes Factor analyses, which offer information about the strength of support for true null hypotheses, were computed using the or package. Rank-based non-parametric tests including two-sided Wilcoxon signed rank tests and two-sided Wilcoxon rank sum tests were used for all analyses, due to the presence of non-normality in the data due to some outliers. No mathematical corrections were made for multiple comparisons in the statistical analysis.

Design matrix

The full design matrix used in all experiments consisted of 48 unique trials, with 16 possible Gain trials, 16 Mixed trials, and 16 Loss trials. Gain trials consisted of a choice between a certain gain and a gamble with equal probabilities of a larger gain or zero. There were two certain point amounts (+25, +50) and the risky gain amount was determined using eight multipliers on the certain amount (1.64, 1.76, 1.88, 2.0, 2.2, 2.5, 3.0, 4.0). The Loss trials, which involved a certain loss and a gamble with zero and a larger loss, also used two certain point amounts (−25, −50) and the same eight multipliers as used for gain trials. The Mixed trials consisted of choices between a certain zero outcome and a gamble with equal probabilities of a gain or a loss. There were two gamble gain amounts featured in Mixed trials (+40, +80), and gamble loss amounts were determined by eight multipliers on these gain amounts (0.2, 0.4, 0.55, 0.67, 0.8, 1.0, 1.33, 2.0).

Experiment 1 featured 192 trials. For each participant, 16 Gain, 16 Mixed, and 16 Loss trials were used to generate a set of 48 unique choice trials. These choice trials were repeated four times in random order such that each choice trial was presented once as a rare sequence trial and three times as a common sequence trial, such that rare and common trials were matched. Experiments 2–7 featured 120 trials. For each participant, Gain, Mixed, and Loss trials are drawn from the unique 16 Gain, 16 Mixed, and 16 Loss trials, generating sets of 30 choice trials, each with the constraint that all 8 unique multipliers for every trial type is represented at least once. Four such sets of 30-choice trials were generated, with one set assigned as rare sequences and three assigned as common sequences. The 120 trials were then pseudo-randomly shuffled, ensuring that every 12 trials included exactly 3 rare sequences and 9 common sequences.

Pre-registration

Experiment 7 was pre-registered on OSF on September 2nd, 2023 (https://doi.org/10.17605/OSF.IO/PA75Q). The original pre-registered hypotheses and exploratory analyses are included in the Supplementary Information.

Parametric decision model based on prospect theory

Prospect theory is a widely used model for economic decision making under risk[21]. The theory describes subjective risk preferences as being composed of two phenomena: the diminishing subjective utility of increasing rewards resulting in risk aversion in gains and risk taking in losses, and the tendency to weigh potential losses more heavily than potential gains (i.e., loss aversion). Within the Prospect Theory framework, the subjective utility of risk taking can be quantified with the following equation:

$$\begin{aligned} \text{Utility}_{\text{risky}} &= 0.5\left(\text{V}_{\text{gain}}\right)^{\alpha_{\text{gain}}} - 0.5\lambda\left(-\text{V}_{\text{loss}}\right)^{\alpha_{\text{loss}}} \\ \text{Utility}_{\text{certain}} &= \left(V_{\text{certain}}\right)^{\alpha_{\text{gain}}} \; if \; V_{\text{certain}} \geq 0 \\ \text{Utility}_{\text{certain}} &= -\lambda\left(-V_{\text{certain}}\right)^{\alpha_{\text{loss}}} \; if \; V_{\text{certain}} < 0 \end{aligned} \tag{4}$$

where V_{gain} and V_{loss} are the values of the potential loss and gain from the risky option, α_{gain} captures the degree of risk aversion for gain trials, α_{loss} captures the degree of risk aversion for losses, and λ is the relative weighting of gains and losses to capture loss aversion. The softmax function was used to determine choice probabilities from subjective utilities:

$$P_{\text{risky}}(t) = \frac{1}{1 + \exp(-[\mu(\text{Utility}_{\text{risky}}(t) - \text{Utility}_{\text{safe}}(t))])} \tag{5}$$

Parameters for all computational models discussed in the paper were estimated using Maximum Likelihood estimation. In models that include difference parameters such as the winning Risky Bias Perseveration Difference model, the Maximum Likelihood estimation procedure included constraints to prevent the sum of parameters and difference parameters exceeding the individual parameter constraints.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The anonymized raw data of all experiments are openly available on OSF at: https://doi.org/10.17605/OSF.IO/PA75Q.

Code availability

Analyses were performed using MATLAB Version: 9.13.0.2049777 (R2022b) and the following packages: Optimization Toolbox Version 9.4 (R2022b) and Statistics and Machine Learning Toolbox Version 12.4 (R2022b). Bayes Factor statistics were computed using the bayesFactor package (v2.3.0), available at: https://doi.org/10.5281/zenodo.7006300. All code to recreate the main findings of this study are available at: https://doi.org/10.17605/OSF.IO/PA75Q.

References

1. Kakade, S. & Dayan, P. Dopamine: generalization and bonuses. *Neural Netw.* **15**, 549–559 (2002).
2. Clewett, D., DuBrow, S. & Davachi, L. Transcending time in the brain: how event memories are constructed from experience. *Hippocampus* **29**, 162–183 (2019).
3. Swallow, K. M. & Jiang, Y. V. The attentional boost effect: transient increases in attention to one task enhance performance in a second task. *Cognition* **115**, 118–132 (2010).
4. Tobler, P. N., Fiorillo, C. D. & Schultz, W. Adaptive coding of reward value by dopamine neurons. *Science* **307**, 1642–1645 (2005).
5. Parkhurst, D., Law, K. & Niebur, E. Modeling the role of salience in the allocation of overt visual attention. *Vis. Res.* **42**, 107–123 (2002).
6. Hakim, N., Feldmann-Wüstefeld, T., Awh, E. & Vogel, E. K. Perturbing neural representations of working memory with task-irrelevant interruption. *J. Cogn. Neurosci.* **32**, 558–569 (2020).
7. Li, X. & Camerer, C. F. Predictable effects of visual salience in experimental decisions and games. *Q. J. Econ.* **137**, 1849–1900 (2022).
8. Schröger, E. & Wolff, C. Behavioral and electrophysiological effects of task-irrelevant sound change: a new distraction paradigm. *Cogn. Brain Res.* **7**, 71–87 (1998).
9. Garrido, M. I., Dolan, R. J. & Sahani, M. Surprise leads to noisier perceptual decisions. *i-Perception* **2**, 112–120 (2011).
10. Dantzer, R. Behavioral, physiological and functional aspects of stereotyped behavior: a review and a re-interpretation. *J. Anim. Sci.* **62**, 1776–1786 (1986).
11. Randrup, A. & Munkvad, I. Pharmacology and physiology of stereotyped behavior. *J. Psychiatr. Res.* **11**, 1–10 (1974).
12. Tomov, M. S., Truong, V. Q., Hundia, R. A. & Gershman, S. J. Dissociable neural correlates of uncertainty underlie different exploration strategies. *Nat. Commun.* **11**, 2371 (2020).
13. Gershman, S. J. Uncertainty and exploration. *Decision* **6**, 277–286 (2019).
14. Kobayashi, K. & Hsu, M. Neural mechanisms of updating under reducible and irreducible uncertainty. *J. Neurosci.* **37**, 6972–6982 (2017).
15. Rutledge, R. B., Skandali, N., Dayan, P. & Dolan, R. J. A computational and neural model of momentary subjective well-being. *Proc. Natl Acad. Sci. USA* **111**, 12252–12257 (2014).
16. Tom, S. M., Fox, C. R., Trepel, C. & Poldrack, R. A. The neural basis of loss aversion in decision-making under risk. *Science* **315**, 515–518 (2007).
17. Sokol-Hessner, P. et al. Thinking like a trader selectively reduces individuals' loss aversion. *Proc. Natl Acad. Sci. USA* **106**, 5035–5040 (2009).
18. Brooks, H. R. & Sokol-Hessner, P. Quantifying the immediate computational effects of preceding outcomes on subsequent risky choices. *Sci. Rep.* **10**, 9878 (2020).
19. Wacongne, C. et al. Evidence for a hierarchy of predictions and prediction errors in human cortex. *Proc. Natl Acad. Sci. USA* **108**, 20754–20759 (2011).
20. Chennu, S. et al. Expectation and attention in hierarchical auditory prediction. *J. Neurosci.* **33**, 11194–11205 (2013).
21. Kahneman, D. & Tversky, A. Prospect theory: an analysis of decision under risk. *Econometrica* **47**, 263–291 (1979).
22. Peterson, J. C., Bourgin, D. D., Agrawal, M., Reichman, D. & Griffiths, T. L. Using large-scale experiments and machine learning to discover theories of human decision-making. *Science* **372**, 1209–1214 (2021).
23. Tversky, A. & Kahneman, D. Advances in prospect theory: cumulative representation of uncertainty. *J. Risk Uncertain.* **5**, 297–323 (1992).
24. Fennema, H. & Wakker, P. Original and cumulative prospect theory: a discussion of empirical differences. *J. Behav. Decis. Mak.* **10**, 53–64 (1997).
25. Edwards, W. The theory of decision making. *Psychol. Bull.* **51**, 380–417 (1954).
26. Harless, D. W. & Camerer, C. F. The predictive utility of generalized expected utility theories. *Econometrica* **62**, 1251–1289 (1994).
27. Von Neumann, J. & Morgenstern, O. *Theory of Games and Economic Behavior* (Princeton University Press, 1944).
28. Barto, A., Mirolli, M. & Baldassarre, G. Novelty or surprise? *Front. Psychol.* **4**, 907 (2013).

29. Todorovic, A., van Ede, F., Maris, E. & de Lange, F. P. Prior expectation mediates neural adaptation to repeated sounds in the auditory cortex: an MEG study. *J. Neurosci.* **31**, 9118–9123 (2011).
30. Modirshanechi, A., Becker, S., Brea, J. & Gerstner, W. Surprise and novelty in the brain. *Curr. Opin. Neurobiol.* **82**, 102758 (2023).
31. Bekinschtein, T. A. et al. Neural signature of the conscious processing of auditory regularities. *Proc. Natl Acad. Sci. USA* **106**, 1672–1677 (2009).
32. Lecaignard, F., Bertrand, O., Caclin, A. & Mattout, J. Neurocomputational underpinnings of expected surprise. *J. Neurosci.* **42**, 474–486 (2022).
33. Chernoff, C. S., Hynes, T. J. & Winstanley, C. A. Noradrenergic contributions to cue-driven risk-taking and impulsivity. *Psychopharmacology* **238**, 1765–1779 (2021).
34. Stalnaker, T. A. et al. Dopamine neuron ensembles signal the content of sensory prediction errors. *eLife* **8**, e49315 (2019).
35. Valdés-Baizabal, C., Carbajal, G. V., Pérez-González, D. & Malmierca, M. S. Dopamine modulates subcortical responses to surprising sounds. *PLoS Biol.* **18**, e3000744 (2020).
36. Sharpe, M. J. et al. Dopamine transients are sufficient and necessary for acquisition of model-based associations. *Nat. Neurosci.* **20**, 735–742 (2017).
37. Suarez, J. A., Howard, J. D., Schoenbaum, G. & Kahnt, T. Sensory prediction errors in the human midbrain signal identity violations independent of perceptual distance. *eLife* **8**, e43962 (2019).
38. Gershman, S. J. & Uchida, N. Believing in dopamine. *Nat. Rev. Neurosci.* **20**, 703–714 (2019).
39. Schultz, W. Behavioral dopamine signals. *Trends Neurosci.* **30**, 203–210 (2007).
40. Chew, B. et al. Endogenous fluctuations in the dopaminergic midbrain drive behavioral choice variability. *Proc. Natl Acad. Sci. USA* **116**, 18732–18737 (2019).
41. Soutschek, A. et al. Dopaminergic D1 receptor stimulation affects effort and risk preferences. *Biol. Psychiatry* **87**, 678–685 (2020).
42. Rigoli, F., Rutledge, R. B., Chew, B., Ousdal, O. T., Dayan, P. & Dolan, R. J. Dopamine increases a value-independent gambling propensity. *Neuropsychopharmacology* **41**, 2658–2667 (2016).
43. Rutledge, R. B., Skandali, N., Dayan, P. & Dolan, R. J. Dopaminergic modulation of decision making and subjective well-being. *J. Neurosci.* **35**, 9811–9822 (2015).
44. Jaskir, A. & Frank, M. J. On the normative advantages of dopamine and striatal opponency for learning and choice. *eLife* **12**, e85107 (2023).
45. Rutledge, R. B. et al. Dopaminergic drugs modulate learning rates and perseveration in Parkinson's patients in a dynamic foraging task. *J. Neurosci.* **29**, 15104–15114 (2009).
46. Kanen, J. W., Ersche, K. D., Fineberg, N. A., Robbins, T. W. & Cardinal, R. N. Computational modelling reveals contrasting effects on reinforcement learning and cognitive flexibility in stimulant use disorder and obsessive-compulsive disorder: remediating effects of dopaminergic D2/3 receptor agents. *Psychopharmacology* **236**, 2337–2358 (2019).
47. Bari, B. A. & Gershman, S. J. Undermatching is a consequence of policy compression. *J. Neurosci.* **43**, 447–457 (2023).
48. Voon, V. et al. Dopamine agonists and risk: impulse control disorders in Parkinson's disease. *Brain* **134**, 1438–1446 (2011).
49. Brown, H. R. et al. Crowdsourcing for cognitive science—the utility of smartphones. *PLoS ONE* **9**, e100662 (2014).
50. Timmer, M. H. M., Sescousse, G., Esselink, R. A. J., Piray, P. & Cools, R. Mechanisms underlying dopamine-induced risky choice in Parkinson's disease with and without depression (history). *Comput. Psychiatr.* **2**, 11–27 (2018).
51. Wetzel, N., Schröger, E. & Widmann, A. The dissociation between the P3a event-related potential and behavioral distraction. *Psychophysiology* **50**, 920–930 (2013).
52. Dalton, P. & Lavie, N. Auditory attentional capture: effects of singleton distractor sounds. *J. Exp. Psychol. Hum. Percept. Perform.* **30**, 180–93 (2004).
53. Jenni, N. L., Larkin, J. D. & Floresco, S. B. Prefrontal dopamine D1 and D2 receptors regulate dissociable aspects of decision making via distinct ventral striatal and amygdalar circuits. *J. Neurosci.* **37**, 6200–6213 (2017).
54. Segalowitz, S. J. & Barnes, K. L. The reliability of ERP components in the auditory oddball paradigm. *Psychophysiology* **30**, 451–459 (1993).
55. Ergen, M., Marbach, S., Brand, A., Başar-Eroğlu, C. & Demiralp, T. P3 and delta band responses in visual oddball paradigm in schizophrenia. *Neurosci. Lett.* **440**, 304–308 (2008).
56. Ostwald, D. et al. Evidence for neural encoding of Bayesian surprise in human somatosensation. *NeuroImage* **62**, 177–188 (2012).
57. Restuccia, D. et al. Somatosensory mismatch negativity in healthy children. *Dev. Med. Child Neurol.* **51**, 991–998 (2009).
58. Gershman, S. J. & Tzovaras, B. G. Dopaminergic genes are associated with both directed and random exploration. *Neuropsychologia* **120**, 97–104 (2018).
59. Thompson, W. R. On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. *Biometrika* **25**, 285–294 (1933).
60. Daw, N. D., O'Doherty, J. P., Dayan, P., Seymour, B. & Dolan, R. J. Cortical substrates for exploratory decisions in humans. *Nature* **441**, 876–879 (2006).
61. Dubois, M. et al. Human complex exploration strategies are enriched by noradrenaline-modulated heuristics. *eLife* **10**, e59907 (2021).
62. Speekenbrink, M. & Konstantinidis, E. Uncertainty and exploration in a restless bandit problem. *Top. Cogn. Sci.* **7**, 351–367 (2015).
63. Bornstein, A. M., Khaw, M. W., Shohamy, D. & Daw, N. D. Reminders of past choices bias decisions for reward in humans. *Nat. Commun.* **8**, 15958 (2017).
64. Decker, J. H., Otto, A. R., Daw, N. D. & Hartley, C. A. From creatures of habit to goal-directed learners: tracking the developmental emergence of model-based reinforcement learning. *Psychol. Sci.* **27**, 848–858 (2016).
65. Cockburn, J., Man, V., Cunningham, W. & O'Doherty, J. P. Novelty and uncertainty regulate the balance between exploration and exploitation through distinct mechanisms in the human brain. *Neuron* **110**, 2691–2702 (2022).
66. Lee, S. W., Shimojo, S. & O'Doherty, J. P. Neural computations underlying arbitration between model-based and model-free learning. *Neuron* **81**, 687–699 (2014).
67. Pisupati, S., Chartarifsky-Lynn, L., Khanal, A. & Churchland, A. K. Lapses in perceptual decisions reflect exploration. *eLife* **10**, e55490 (2021).
68. Payzan-LeNestour, E., Pradier, L., Doran, J., Nave, G. & Balleine, B. Impact of ambient sound on risk perception in humans: neuroeconomic investigations. *Sci. Rep.* **11**, 5392 (2021).
69. Woods, K. J. P., Siegel, M. H., Traer, J. & McDermott, J. H. Headphone screening to facilitate web-based auditory experiments. *Atten. Percept. Psychophys.* **79**, 2064–2072 (2017).
70. Anwyl-Irvine, A. L., Massonnié, J., Flitton, A., Kirkham, N. & Evershed, J. K. Gorilla in our midst: an online behavioral experiment builder. *Behav. Res. Methods* **52**, 388–407 (2020).

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Author contributions

G.W.F. and R.B.R. designed research; G.W.F. performed research; G.W.F. analyzed data; G.W.F. and R.B.R. wrote the paper.

Competing interests

G.W.F. declares no competing interests. R.B.R. holds equity in Maia.

Additional information

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